

CalmGlobal KPI Platform

Operations & Deployment Guide — updated 9 October 2026 (v2)

1. What it is

A performance-management web application for CalmGlobal: KPI tracking with alerts, staff performance appraisals (self-assessment → unit-head review → MD review → final grade), departments, notifications, audit logs, PDF/Excel exports, and work-item sync from Azure DevOps / Jira / Asana.

Component	Technology	Where it runs
Frontend	React 18 + Vite + Tailwind 4 (static build)	IIS site KPI-Web → <code>C:\apps\kpi-platform\frontend\dist</code>
Backend API	Node.js / Express 5 + Prisma ORM	IIS sub-application /api (HttpPlatformHandler) → <code>C:\apps\kpi-platform\backend</code>
Database	PostgreSQL 16, database <code>kpi_platform</code>	Same VPS, <code>localhost:5432</code>
Email	MailerSend SMTP (calmglobal.com domain)	Configured in <code>backend\.env</code>

2. Addresses & access

What	Address
Live application	https://kpi.calmglobal.com (HTTP redirects to HTTPS automatically)
API health check	https://kpi.calmglobal.com/api/health → <code>{"status":"ok", ...}</code>
Server (VPS)	OVH Windows Server, 158.69.198.49 — access via RDP
Code — primary	Azure DevOps: dev.azure.com/calmglobalkoko → project CALMGLOBAL KPI
Code — mirror	GitHub: github.com/iamlekside2/kpi-platform
Pipeline	Azure DevOps → Pipelines → CALMGLOBAL KPI
DNS	GoDaddy (calmglobal.com zone): A record <code>kpi</code> → <code>158.69.198.49</code>
SSL certificate	Let's Encrypt via win-acme; renews automatically (scheduled task on the VPS)

3. How to deploy a change (the normal way)

Everything is automated through the Azure DevOps pipeline and a self-hosted agent (**KPI-VPS**, pool **calm**) that runs on the VPS itself:

1. Edit code locally (repo: D:\kpi-platform on the work laptop)
2. `git add .`
`git commit -m "describe the change"`
`git push azure main` (and optionally: `git push origin main`)
3. Done. The pipeline builds the frontend, installs backend deps, syncs the database schema (prisma db push), copies files into C:\apps\kpi-platform, restarts the backend and health-checks it.

Watch progress: Azure DevOps → Pipelines → latest run (~6–8 minutes). Green = deployed and verified.

No server access is needed for code changes. RDP to the VPS is only required for infrastructure work: new secrets in `.env`, certificates for new domains, IIS changes.

4. Users, roles & registration

Action	How
Create an account	https://kpi.calmglobal.com/register → onboarding creates your organisation and makes you its admin . One self-registration per company domain : once the first person from a company registered, colleagues with the same email domain are blocked and must be added by the admin (public providers like Gmail are exempt).
Add staff	Staff page (as admin) — new members get an invite email with temporary password <code>Welcome@123</code> (they must change it after first login: Settings → Security)
Roles	admin (HR: full org control, admin comments) · lead (Unit Head: team dashboard + Section 2 reviews) · md (sees everything, writes the final MD comments/score) · chairman (view-only access to everything) · member (self-assessments, own data)
Dashboards	Role-aware: lead → "My Team" (pending reviews, team ADO stats) · md/chairman → "Executive Overview" (completion %, avg score, grade/department charts, staff performance table) · admin/member → org dashboard
Azure DevOps identity	If someone's ADO account uses a different email (e.g. personal Gmail), they set it themselves: Profile → "Azure DevOps Email" — the staff sync matches either address
Forgot password	Login page → "Forgot password?" — emails a reset link valid for 15 minutes

5. Email (configured & verified working)

Sender: `kpi@calmglobal.com` via MailerSend SMTP (`smtp.mailersend.net:587`, calmglobal.com domain verified — same account as the RealCousins project). Emails sent: welcome on registration, staff invites, password reset, password-changed confirmation, and KPI alert notifications from the sync module.

5b. Azure DevOps integration — what it captures

Admin connects it in Settings → Integrations (org URL `https://dev.azure.com/calmglobal` + a PAT with Work Items Read scope). Two sync types, both also run hourly by the scheduler:

- **Sync KPIs** — auto-creates and updates 8 org KPIs (last 30 days unless noted): Tasks Completed, Open Items, Blocked Count, Items Created, Bugs Created, Bugs Fixed, Open Bugs (live snapshot), Story Points Completed.
- **Sync Staff Tasks** — attaches each person's work items to their profile (used in appraisals and the team/executive dashboards); matches by login email or the per-user "Azure DevOps Email".

The state mapping understands the CalmGlobal custom board states (`Done Deployed` , `In Sprint` , `Confirmed Awaiting QA/Release` → open, `Pending Issues` → blocked, etc.). Sync results and errors: Settings → Sync History.

Settings live only in `C:\apps\kpi-platform\backend\.env` (`SMTP_*` , `EMAIL_FROM`). After changing them:
`C:\Windows\System32\inetsrv\appcmd.exe recycle apppool /apppool.name:kpi-api`

6. Server layout (VPS 158.69.198.49)

Item	Value
App folder	<code>C:\apps\kpi-platform</code> (git clone; the pipeline overwrites <code>backend/</code> and <code>frontend/dist</code>)
IIS site	KPI-Web — bindings: http/https <code>kpi.calmglobal.com</code> ; root = frontend <code>dist</code> ; sub-app <code>/api</code> = backend
App pools	kpi-web (static) · kpi-api (No Managed Code, AlwaysRunning , idle timeout 0 — keeps the hourly sync scheduler alive; do not change)
Backend secrets	<code>C:\apps\kpi-platform\backend\.env</code> — DATABASE_URL, JWT secrets, FRONTEND_URL, SMTP. Never in git.
Backend logs	<code>C:\apps\kpi-platform\backend\logs\node_*.log</code> (Node stdout/stderr per process)
Build agent	<code>C:\azagent</code> — service <code>vstsagent.calmglobalkoko.calm.KPI-VPS</code> , runs as SYSTEM
win-acme (SSL)	<code>C:\tools\win-acme</code> — renewal runs via Windows scheduled task
Database	PostgreSQL 16 (<code>C:\Program Files\PostgreSQL\16</code>), shared with other apps (dossiro etc.); connect as the user set in DATABASE_URL

7. Runbook — common operations (on the VPS, admin PowerShell)

Restart the backend

```
C:\Windows\System32\inetsrv\appcmd.exe recycle apppool /apppool.name:kpi-api
```

View recent backend logs

```
$f = Get-ChildItem C:\apps\kpi-platform\backend\logs\node*.log | Sort-Object LastWriteTime -Descending | Select-Object -First 1; Get-Content $f.FullName -Tail 60
```

Force a database schema sync (after pulling schema changes manually)

```
cd C:\apps\kpi-platform\backend; npx prisma db push
```

Helper scripts in the repo (C:\apps\kpi-platform\deploy)

Script	Purpose
<code>setup-server.ps1</code>	Full one-time server setup (already done — keep for rebuilds)
<code>pipeline-deploy.ps1</code>	What the pipeline executes on each push (do not run manually unless debugging)
<code>fix-db.ps1</code>	Reset/verify DB credentials in .env, create DB, sync, restart
<code>reset-pg-password.ps1</code>	Recover the postgres superuser password (temporary localhost trust)
<code>get-ssl.ps1</code> / <code>bind-https.ps1</code>	Issue the Let's Encrypt cert / attach it to the IIS binding
<code>diagnose-api.ps1</code>	One-shot health report: pools, sites, node processes, logs, IIS errors
<code>setup-agent.ps1</code>	Install the Azure DevOps build agent (already done)

8. Troubleshooting — known issues & fixes

Symptom	Cause	Fix
/api returns empty 403 , backend logs empty	The frontend's SPA rewrite rule inherited into the /api sub-app	Backend <code>web.config</code> must keep <code><rewrite><rules><clear/></rules></rewrite></code> (already in <code>deploy/backend.web.config</code>)
"Not allowed by CORS" in logs; login fails	Site used over plain HTTP	Use HTTPS (HTTP now auto-redirects). CORS allows only <code>FRONTEND_URL</code> in production
500 with Prisma P2022 "column does not exist"	DB schema behind <code>schema.prisma</code> (repo's migration history is incomplete)	<code>npx prisma db push</code> then recycle <code>kpi-api</code> — the pipeline now does this on every deploy
Logged out after page refresh	Refresh cookie is <code>Secure</code> — only works over HTTPS	Use HTTPS (enforced now)
Pipeline stuck "waiting for an agent"	Agent offline or pool mismatch	Check service <code>vstsagent.calmglobalkoko.calm.KPI-VPS</code> is running; YAML pool must be <code>calm</code>
Emails not arriving	SMTP settings / sender identity	Check <code>[SMTP]</code> lines in node log; verify sender domain in MailerSend; check spam folder
Password reset "fails" silently	No email provider configured	Ensure <code>SMTP_*</code> set in <code>.env</code> and restart pool

9. Security notes

Rotate when convenient (values were exchanged in chat during setup): the PostgreSQL passwords (both the `postgres` and `dossiro` users) and the MailerSend SMTP password. After rotating, update `backend\.env` here (and RealCousins' `.env` for SMTP) and recycle the app pools.

- JWT secrets are random, generated at setup, stored only in `backend\.env`.
- Staff invites use fixed temp password `Welcome@123` — consider changing to random per-invite.
- Public registration is open (see §4). Decide: restrict, or keep open for multi-tenant use.
- The Brevo fallback in `mailer.js` has a hardcoded personal Gmail default sender — harmless while Brevo is unused, but worth cleaning up.

10. Open product decisions

1. ~~Registration lockdown~~ — **done**: one self-registration per company domain; colleagues join by admin invite.
2. ~~Executive visibility~~ — **done**: MD and Chairman roles with the Executive Overview dashboard.
3. **Landing page** — the root URL goes straight to login; add a marketing/landing page if the product will be shown externally. Still open.